

XUEJIAN(JACOB) SHEN

xuejianshen@fas.harvard.edu ♦ [Personal Site](#) ♦ ORCID [0000-0002-6196-823X](#)

Center for Astrophysics, Harvard University

[last updated on July 29, 2026]

EDUCATION

California Institute of Technology Ph.D., Physics; Advisor: Prof. Philip Hopkins Thesis: Cosmic structure and galaxy formation in alternative dark matter	Sept 2018 - June 2023
Peking University B.S., Physics	Sept 2014 - June 2018

EMPLOYMENT

ITC Fellow Center for Astrophysics, Harvard University	Jul 2026 - present
Postdoctoral Fellow Kavli Institute for Astrophysics and Space Research Massachusetts Institute of Technology	Oct 2023 - Jul 2026
SubMIT Project Team Department of Physics, Massachusetts Institute of Technology	Oct 2023 - Jul 2026
Graduate Teaching & Research Assistant California Institute of Technology	Jan 2019 - June 2023

RESEARCH INTERESTS

Cosmological simulations of galaxy formation; The nature of dark matter; Formation and evolution of galaxies and supermassive black holes at high redshift
UAT Keywords: galaxy formation (595) – galaxy evolution (594) – dark matter (353)

SUMMARY OF PUBLICATIONS

Total publications: **97**
First-author publications: **14**; From supervised students: **11**
Metrics: **>3000 citations**, **>1100 first/corresponding-author citations**, **h-index: 31**
(see attached publication list for details)

AWARDS & FELLOWSHIPS

Outstanding UROP Mentor Award	Massachusetts Institute of Technology, 2026
ITC Fellowship	CfA, Harvard University, 2025
Neil Gehrels Prize Fellowship (declined)	University of Maryland, 2023
KIPAC Fellowship (declined)	Stanford University, 2023
TCCAP Fellowship (declined)	University of Texas at Austin, 2023
James A. Cullen Memorial Fellowship	California Institute of Technology, 2022
Honored Graduate	Peking University, 2018
Robin Li Fellowship	Peking University, 2017
Meritorious Award	Mathematical Contest in Modeling, 2017
Weiming Scholar Fellowship	Peking University, 2015, 2016
HaiLiang Fellowship	Peking University, 2016
GuangHua Fellowship	Peking University, 2015

RESEARCH GRANTS

As Principal Investigator (PI)

ITC Fellowship	300000	07/2026 – 07/2029
JWST-AR-Cycle3 Theory (04814)	\$329597.05	10/2024 – 10/2027

As Co-Investigator

JWST-GO-Cycle4 (07404, PI Naidu)	49.3 hours	10/2025 – 10/2027
SuperMUC-NG at LRZ (PI Puchwein)	33M core hours	09/2025 – 09/2027
DiRAC THESAN-MultiZoom (PI McClymont)	18.4M core hours	01/2026 – 01/2028
THESAN-CGM refinement (PI Pruto)	5.2M core hours	01/2026 – 01/2028
LUMINA/LUMINA-ALT (PI Zier)	>150M core hours	01/2025 – 01/2028

TEACHING & ADVISING

Caltech Graduate Teaching Assistant, Computational Physics Lab Ph20/21/22
Lectures, MIT Physics 8.902 (2023, 2024)

Undergraduate & Master Students:

Gabriel Aguiar (Caltech, 2021-2022), Eitan Rapaport (Caltech, 2022-2023), Evan Erickson (MIT, 2023-), Yongao Hu (MIT, 2024), Hui Wang (MIT, 2023-2024), Vinh Tran (MIT, 2023-), Aidan Leonard (MIT, 2023-2024), Zihao Wang (MIT/PKU, 2023-), Charline Shen (Harvard, 2023-2025), Qiaorong Yu (Oxford, 2025), Xiaoqing Sun (MIT, 2024), Aliaksandr Melnichenka (Berea College, 2025), Hyun Song (Harvard, 2026-), Ye He (NJU, 2026-), Jiahui Xie (Caltech/Harvard, 2026-), Chuiyang Kong (Brown/Harvard, 2026-)

Graduate Students: Rongrong Liu (Harvard, 2025-), Darkhan Shadykul (MIT, 2026-)

SYNERGISTIC ACTIVITIES

Professional Services

- Journal Referee for Monthly Notices of the Royal Astronomical Society (MNRAS, since 2020), MNRAS Letters (since 2023), The Astrophysical Journal (ApJ, since 2022), ApJ Letters (since 2022), Astronomy & Astrophysics (since 2025), Scientific Reports (Since 2023), Physics Review

Letters (PRL, since 2024), Physics Review D (PRD, since 2023)

- Oscii Bascii (& Grad student representative) of Theoretical AstroPhysics Including Relativity and Cosmology (TAPIR, 2019-2022), Caltech
- Local Organizing Committee (LOC) for the Galaxy Formation and Evolution in Southern California (GalFRESKA) workshop (2022)
- MIT Physics Basic Computing Services “SubMIT” Project Team Member (since 2023)
- Organizer for MIT Physics Basic Computing Services User Meetings (since 2023)

Outreach

- Speaker at Caltech Stargazing Lecture Series (2023)
- Organizer of Stargazing and Outreach Activities (2015-2018), Peking University

SELECTED TALKS

Invited Talks

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| - Learning the Universe at Cosmic Frontier | Penn State University, 04/2026 |
| - Learning the Universe at Cosmic Frontier | Westlake University, 01/2026 |
| - Galaxy formation models at the cosmic dawn: Insights from JWST | Shanghai Jiaotong University, 01/2026 |
| - Learning the Universe at Cosmic Frontier | Nanjing University, 12/2025 |
| - Learning the Universe at Cosmic Frontier | Fudan University, 12/2025 |
| - Cosmic Dawn as a Laboratory for New Physics | Shanghai Jiaotong University, 12/2025 |
| - Cosmic Dawn as a Laboratory for New Physics | HEP seminar, Harvard University, 11/2025 |
| - Cosmic Dawn as a portal to New Physics | Boston University, 10/2025 |
| - Cosmic Dawn as a portal to New Physics | Nanjing University, 09/2025 |
| - Cosmic Dawn as a portal to New Physics | Peking University, 09/2025 |
| - Cosmic Dawn as a portal to New Physics | Tsinghua University, 09/2025 |
| - Alternative dark matter in the THESAN simulations | DREAMS collaboration meeting, CCA, 05/2025 |
| - Challenges to galaxy formation models at the cosmic frontier | Yale University, 04/2025 |
| - Challenges to galaxy formation models at the cosmic frontier | CCA, Flatiron Institute, 10/2024 |
| - Challenges to galaxy formation models at the cosmic frontier | Zhejiang University, 09/2024 |
| - Challenges to galaxy formation models at the cosmic frontier | Univ. of Texas Austin, 08/2024 |
| - Challenges to galaxy formation models at the cosmic frontier | Univ. of Cambridge, 06/2024 |
| - The implication of UV variability for the bright galaxy abundance at cosmic dawn | Tsinghua University, 09/2023 |
| - The implication of UV variability for the bright galaxy abundance at cosmic dawn | KIAA seminar, Peking Univ., 09/2023 |
| - Dark Matter: Elusive fibers of the Universe | Stargazing Lecture, Caltech, 05/2023 |
| - New aspects about dark matter models beyond CDM | KIPAC tea talk, Stanford, 02/2023 |
| - Alternative dark matter and structure formation | Obs. Cosmo. seminar, Caltech, 11/2021 |

Contributed Talks

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| - The Lumina Simulation & Puzzles of high-redshift galaxies in the Roman era | Roman Science Meeting, 07/2026 |
| - Basic Computing Service Review: User Support | MIT, 06/2026 |
| - An Introduction to the LUMINA project | Journal Club, MIT, 04/2026 |
| - The cosmic rush hour in a universe with early dark energy | Tokyo, 03/2026 |

- The cosmic rush hour in a universe with early dark energy Aspen, 03/2026
- The cosmic rush hour in a universe with early dark energy ITC Luncheon, 02/2026
- The cosmic rush hour in a universe with early dark energy Napa Valley, 01/2026
- Cosmic Dawn as a portal to New Physics FOLIAGE meeting, 11/2025
- Quasar luminosity functions and clustering in LUMINA Thesan workshop, Harvard CfA, 08/2025
- The rich astrophysical phenomenology in a dissipative dark sector SIDM workshop, Valencia, 06/2025
- Flash talk: Supermassive black hole seeds from dark matter CFC meeting, Austin, 05/2025
- Basic Computing Service Review: OpenMPI and Data Transfer MIT, 05/2025
- First results from the THESAN-ZOOM simulations Journal Club, MIT, 03/2025
- Star-formation efficiencies in high-redshift galaxies learned from THESAN-ZOOM Harvard CfA, 03/2025
- Early dark energy as a unified solution to the Hubble tension and early galaxy puzzles raised by JWST KITP conference, 08/2024
- Insights on efficiency and variability of star-formation from radiation-hydro simulations Santa Cruz Galaxy Formation Workshop, 08/2024
- Basic Computing Service Review: Customizations and Community Engagement MIT, 08/2024
- Challenges to galaxy formation models at the cosmic frontier Journal Club, MIT, 04/2024
- Challenges to galaxy formation models at high redshifts University of Vienna, 02/2024
- Galaxy sizes during the epoch of reionization “Building Galaxies from Scratch” Harvard CfA, 02/2024
- The implication of UV variability for the bright galaxy abundance at cosmic dawn ITC luncheon, Harvard CfA, 10/2023
- Galaxy in the epoch of reionization in alternative dark matter UCLA DM meeting, 03/2023
- Alternative dark matter in galaxy formation Dark Cosmos seminar, Princeton, 10/2022
- Alternative dark matter in galaxy formation Brown Bag lunch talk, MIT, 10/2022
- Alternative dark matter in galaxy formation Harvard CfA, 10/2022
- Alternative dark matter in galaxy formation FIRE seminar, 10/2022
- Dwarf galaxies in dissipative dark matter GalFRESKA workshop, 09/2022
- High-redshift predictions from the Illustris-TNG simulations MIT, 08/2019
- High-redshift predictions from the Illustris-TNG simulations Caltech, 06/2019

SELECTED PRESS COVERAGE

Kavli Foundation News: "Early" Dark Energy Could Explain Anomalous Galaxies and the Universe's Disjointed Expansion Rate
Univ. of Cambridge News: Dark energy in the early universe could solve two major problems in cosmology
UT Austin News: Early Dark Energy Could Resolve Cosmology's Two Biggest Puzzles
Yahoo News: How dark energy could relieve 'Hubble tension' and galaxy headaches
Phys.org: Early dark energy could resolve cosmology's two biggest puzzles
MirageNews: Early Dark Energy May Solve Cosmology's Top Two Mysteries
MIT News: Early dark energy could resolve cosmology's two biggest puzzles
CCA News: Bursts of Star Formation Explain Mysterious Brightness at Cosmic Dawn
Space.com: Dark matter atoms may form shadowy galaxies with rapid star formation
Northwestern News: Bursts of star formation explain mysterious brightness at cosmic dawn
Phys.org: Dark matter can make dark atoms, say theoretical astrophysicists
Scientific American: JWST's Glimpses of Early Galaxies Could Shed Light on Dark Matter
Astrobites: Forecasting the obscured first few billion years

PUBLICATIONS (FIRST-AUTHOR & PEER-REVIEWED)

See my [NASA ADS bibliography](#) for full information (ORCID **0000-0002-6196-823X**)

1. [X. Shen](#), R. Kannan, E. Puchwein, A. Smith, M. Vogelsberger, et al. The THESAN-ZOOM project: star formation efficiencies in high-redshift galaxies. *MNRAS*, 545(4):staf2119, February 2026
2. [X. Shen](#), M. Vogelsberger, M. Boylan-Kolchin, S. Tacchella, R. P. Naidu. Early galaxies and early dark energy: a unified solution to the hubble tension and puzzles of massive bright galaxies revealed by JWST. *MNRAS*, 533(4):3923–3936, October 2024
3. [X. Shen](#), M. Vogelsberger, J. Borrow, Y. Hu, E. Erickson, et al. The THESAN project: galaxy sizes during the epoch of reionization. *MNRAS*, 534(2):1433–1458, October 2024
4. [X. Shen](#), P. F. Hopkins, L. Necib, F. Jiang, M. Boylan-Kolchin, et al. Dissipative Dark Matter on FIRE. II. Observational Signatures and Constraints from Local Dwarf Galaxies. *ApJ*, 966(1):131, May 2024
5. [X. Shen](#), H. Xiao, P. F. Hopkins, K. M. Zurek. Disruption of Dark Matter Minihalos in the Milky Way Environment: Implications for Axion Miniclusters and Early Matter Domination. *ApJ*, 962(1):9, February 2024
6. [X. Shen](#), J. Borrow, M. Vogelsberger, E. Garaldi, A. Smith, et al. THESAN-HR: galaxies in the Epoch of Reionization in warm dark matter, fuzzy dark matter, and interacting dark matter. *MNRAS*, 527(2):2835–2857, January 2024
7. [X. Shen](#), M. Vogelsberger, M. Boylan-Kolchin, S. Tacchella, R. Kannan. The impact of UV variability on the abundance of bright galaxies at $z \geq 9$. *MNRAS*, 525(3):3254–3261, November 2023
8. [X. Shen](#), T. Brinckmann, D. Rapetti, M. Vogelsberger, A. Mantz, et al. X-ray morphology of cluster-mass haloes in self-interacting dark matter. *MNRAS*, 516(1):1302–1319, October 2022

9. **X. Shen**, M. Vogelsberger, D. Nelson, S. Tacchella, L. Hernquist, et al. High-redshift predictions from IllustrisTNG - III. Infrared luminosity functions, obscured star formation, and dust temperature of high-redshift galaxies. *MNRAS*, 510(4):5560–5578, March 2022
10. **X. Shen**, P. F. Hopkins, L. Necib, F. Jiang, M. Boylan-Kolchin, et al. Dissipative dark matter on FIRE - I. Structural and kinematic properties of dwarf galaxies. *MNRAS*, 506(3):4421–4445, September 2021
11. **X. Shen**, M. Vogelsberger, D. Nelson, A. Pillepich, S. Tacchella, et al. High-redshift JWST predictions from IllustrisTNG: II. Galaxy line and continuum spectral indices and dust attenuation curves. *MNRAS*, 495(4):4747–4768, July 2020
12. **X. Shen**, P. F. Hopkins, C.-A. Faucher-Giguère, D. M. Alexander, G. T. Richards, et al. The bolometric quasar luminosity function at $z = 0-7$. *MNRAS*, 495(3):3252–3275, January 2020

PUBLICATIONS (FIRST-AUTHOR & UNDER REVIEW)

1. **X. Shen**, O. Zier, A. Smith, R. Liu, R. Kannan, et al. The Lumina Project: The Demographics of Active Galactic Nuclei from Quasars to Little Red Dots at $z \geq 3$. *arXiv e-prints*, page arXiv:2605.24112, May 2026
2. **X. Shen**, O. Zier, M. Vogelsberger, M. Boylan-Kolchin, L. Hernquist, et al. The Cosmic Rush Hour: Rapid Formation of Bright, Massive, Disky, Star-Forming Galaxies as Signatures of Early-Universe Physics. *arXiv e-prints*, page arXiv:2509.19427, September 2025

PUBLICATIONS (CO-AUTHORED & PEER-REVIEWED)

[†] indicate publications from supervised students + as corresponding author

1. [†] V. Tran, **X. Shen**, M. Vogelsberger, D. Gilman, S. O’Neil, et al. Novel density profile for isothermal cores of dark matter halos. *Phys. Rev. D*, 113(6):063024, March 2026
2. [†] Z. Wang, **X. Shen**, M. Vogelsberger, H. Li, R. Kannan, et al. The THESAN-ZOOM project: star formation efficiency from giant molecular clouds to galactic scale in high-redshift starbursts. *MNRAS*, 544(3):2675–2697, December 2025
3. [†] Q. S. Yu, S. O’Neil, **X. Shen**, M. Vogelsberger, S. Bose, et al. Measuring the splashback feature: Dependence on halo properties and history. *The Open Journal of Astrophysics*, 8:E164, November 2025
4. [†] V. Tran, **X. Shen**, D. Gilman, M. Vogelsberger, S. O’Neil, et al. Core collapse in resonant self-interacting dark matter across two decades in halo mass. *Phys. Rev. D*, 112(8):083003, October 2025
5. [†] X. Sun, S. O’Neil, **X. Shen**, M. Vogelsberger. The effects of projection on measuring the splashback feature. *The Open Journal of Astrophysics*, 8:100, July 2025
6. [†] S. Roy, **X. Shen**, J. Barron, M. Lisanti, D. Curtin, et al. Aggressively Dissipative Dark Dwarfs: The Effects of Atomic Dark Matter on the Inner Densities of Isolated Dwarf Galaxies. *ApJ*, 982(2):175, April 2025

7. [†] A. Leonard, S. O’Neil, [X. Shen](#), M. Vogelsberger, O. Rosenstein, et al. Varying primordial state fractions in exo- and endothermic SIDM simulations of Milky Way-mass haloes. *MNRAS*, 531(1):1440–1453, June 2024
8. [†] C. Gemmell, S. Roy, [X. Shen](#), D. Curtin, M. Lisanti, et al. Dissipative Dark Substructure: The Consequences of Atomic Dark Matter on Milky Way Analog Subhalos. *ApJ*, 967(1):21, May 2024
9. [†] S. Roy, [X. Shen](#), M. Lisanti, D. Curtin, N. Murray, et al. Simulating Atomic Dark Matter in Milky Way Analogs. *ApJ*, 954(2):L40, September 2023
10. [†] H. Xiao, [X. Shen](#), P. F. Hopkins, K. M. Zurek. SMBH seeds from dissipative dark matter. *J. Cosmol. Astropart. Phys.*, 2021(7):039, July 2021
11. A. M. Garcia, J. C. Rose, P. Torrey, A. Caputo, M. Lisanti, et al., incl. [X. Shen](#). The DREAMS Project: Disentangling the Impact of Halo-to-halo Variance and Baryonic Feedback on Milky Way Dark Matter Density Profiles. *ApJ*, 1002(1):8, May 2026
12. E. Lilie, J. C. Rose, M. Lisanti, A. M. Garcia, P. Torrey, et al., incl. [X. Shen](#). The DREAMS Project: Disentangling the Impact of Halo-to-Halo Variance and Baryonic Feedback on Milky Way Dark Matter Speed Distributions. *ApJ*, 1002(2):168, May 2026
13. W. McClymont, S. Tacchella, A. Smith, R. Kannan, E. Garaldi, et al., incl. [X. Shen](#). The THESAN-ZOOM project: mystery N/O more — uncovering the origin of peculiar chemical abundances and a not-so-fundamental metallicity relation at $3 \lesssim z \lesssim 12$. *MNRAS*, 548(1):stag016, May 2026
14. A. M. Garcia, P. Torrey, A. Bhagwat, [X. Shen](#), M. Vogelsberger, et al. Metallicity Gradients in Modern Cosmological Simulations. II. The Role of Bursty versus Smooth Feedback at High Redshift. *ApJ*, 1001(2):188, April 2026
15. V. Tran, D. Gilman, M. S. Delos, [X. Shen](#), O. Zier, et al. Role of prompt cusps in driving the core collapse of self-interacting dark matter halos. *Phys. Rev. D*, 113(8):083030, April 2026
16. I. S. Sands, P. F. Hopkins, [X. Shen](#), M. Boylan-Kolchin, J. Bullock, et al. Confronting the Diversity Problem: The Limits of Galaxy Rotation Curves as a Tool to Understand Dark Matter Profiles. *ApJ*, 1000(1):127, March 2026
17. T. Nguyen, F. Villaescusa-Navarro, S. Mishra-Sharma, C. Cuesta-Lazaro, P. Torrey, et al., incl. [X. Shen](#). How DREAMS Are Made: Emulating Satellite Galaxy and Subhalo Populations with Diffusion Models and Point Clouds. *ApJ*, 997(2):336, February 2026
18. E. Garaldi, R. Kannan, A. Smith, J. Borrow, M. Vogelsberger, et al., incl. [X. Shen](#). Correction to: The THESAN project: public data release of radiation-hydrodynamic simulations matching reionization-era JWST observations. *MNRAS*, 545(4):staf2196, February 2026
19. G. Pruto, L. Keating, R. Kannan, E. Puchwein, A. Smith, et al., incl. [X. Shen](#). The THESAN-ZOOM project: the hidden neighbours of O I absorbers during reionization. *MNRAS*, 546(2):stag064, February 2026
20. F. Jiang, Z. Jia, H. Zheng, L. C. Ho, K. Inayoshi, et al., incl. [X. Shen](#). Formation of the Little Red Dots from the Core Collapse of Self-interacting Dark Matter Halos. *ApJ*, 996(1):L19, January 2026

21. W. McClymont, S. Tacchella, X. Ji, R. Kannan, R. Maiolino, et al., incl. [X. Shen](#). Overmassive black holes in the early Universe can be explained by gas-rich, dark matter-dominated galaxies. *MNRAS*, 545(1):staf2092, January 2026
22. R. P. Naidu, P. A. Oesch, G. Brammer, A. Weibel, Y. Li, et al., incl. [X. Shen](#). A Cosmic Miracle: A Remarkably Luminous Galaxy at $z_{\text{spec}} = 14.44$ Confirmed with JWST. *The Open Journal of Astrophysics*, 9:56033, January 2026
23. W. McClymont, S. Tacchella, A. Smith, R. Kannan, E. Puchwein, et al., incl. [X. Shen](#). The THESAN-ZOOM project: central starbursts and inside-out quenching govern galaxy sizes in the early Universe. *MNRAS*, 544(2):1732–1747, December 2025
24. B. Costanza, B. Y. Wang, F. Villaescusa-Navarro, A. M. Garcia, J. C. Rose, et al., incl. [X. Shen](#). On the Sensitivity of Different Galaxy Properties to Warm Dark Matter. *ApJ*, 994(1):62, November 2025
25. M. C. Straight, M. Boylan-Kolchin, J. S. Bullock, P. F. Hopkins, [X. Shen](#), et al. Central densities of dark matter haloes in FIRE-2 simulations of low-mass galaxies with cold dark matter and self-interacting dark matter. *MNRAS*, 543(3):1995–2005, November 2025
26. T.-E. Bulichi, O. Zier, A. Smith, M. Vogelsberger, A.-C. Eilers, et al., incl. [X. Shen](#). High-redshift AGN population in radiation-hydrodynamics simulations. *MNRAS*, 544(1):355–371, November 2025
27. O. Zier, R. Kannan, A. Smith, E. Puchwein, M. Vogelsberger, et al., incl. [X. Shen](#). The THESAN-ZOOM project: long-term imprints of external reionization on galaxy evolution. *MNRAS*, 544(1):391–409, November 2025
28. O. Zier, R. Kannan, A. Smith, E. Puchwein, M. Vogelsberger, et al., incl. [X. Shen](#). The THESAN-ZOOM project: Population III star formation continues until the end of reionization. *MNRAS*, 544(1):410–429, November 2025
29. W. McClymont, S. Tacchella, A. Smith, R. Kannan, E. Puchwein, et al., incl. [X. Shen](#). The THESAN-ZOOM project: burst, quench, repeat — unveiling the evolution of high-redshift galaxies along the star-forming main sequence. *MNRAS*, 544(1):513–534, November 2025
30. R. Kannan, E. Puchwein, A. Smith, J. Borrow, E. Garaldi, et al., incl. [X. Shen](#). Introducing the THESAN-ZOOM project: radiation-hydrodynamic simulations of high-redshift galaxies with a multi-phase interstellar medium. *The Open Journal of Astrophysics*, 8:153, October 2025
31. N. Jamieson, A. Smith, M. Neyer, R. Kannan, E. Garaldi, et al., incl. [X. Shen](#). The THESAN project: tracking the expansion and merger histories of ionized bubbles during the Epoch of Reionization. *MNRAS*, 541(2):1088–1105, August 2025
32. D. Narayanan, D. P. Stark, S. L. Finkelstein, P. Torrey, Q. Li, et al., incl. [X. Shen](#). The Ultraviolet Slopes of Early Universe Galaxies: The Impact of Bursty Star Formation, Dust, and Nebular Continuum Emission. *ApJ*, 982(1):7, March 2025
33. R. Feldmann, M. Boylan-Kolchin, J. S. Bullock, O. Çatmabacak, C.-A. Faucher-Giguère, et al., incl. [X. Shen](#). Elevated UV luminosity density at Cosmic Dawn explained by non-evolving, weakly mass-dependent star formation efficiency. *MNRAS*, 536(1):988–1016, January 2025

34. C. Roche, M. McDonald, J. Borrow, M. Vogelsberger, [X. Shen](#), et al. Brightest Cluster Galaxy Offsets in Cold Dark Matter. *The Open Journal of Astrophysics*, 7:65, August 2024
35. V. Tran, D. Gilman, M. Vogelsberger, [X. Shen](#), S. O’Neil, et al. Gravo-thermal catastrophe in resonant self-interacting dark matter models. *Phys. Rev. D*, 110(4):043048, August 2024
36. E. Garaldi, R. Kannan, A. Smith, J. Borrow, M. Vogelsberger, et al., incl. [X. Shen](#). The THESAN project: public data release of radiation-hydrodynamic simulations matching reionization-era JWST observations. *MNRAS*, 530(4):3765–3786, June 2024
37. T. Morishita, M. Stiavelli, R.-R. Chary, M. Trenti, P. Bergamini, et al., incl. [X. Shen](#). Enhanced Subkiloparsec-scale Star Formation: Results from a JWST Size Analysis of 341 Galaxies at $5 < z < 14$. *ApJ*, 963(1):9, March 2024
38. T. K. Waters, C. Peterson, R. Emami, [X. Shen](#), L. Hernquist, et al. Gas Morphology of Milky Way-like Galaxies in the TNG50 Simulation: Signals of Twisting and Stretching. *ApJ*, 961(2):193, February 2024
39. G. Sun, C.-A. Faucher-Giguère, C. C. Hayward, [X. Shen](#). Seen and unseen: bursty star formation and its implications for observations of high-redshift galaxies with JWST. *MNRAS*, 526(2):2665–2672, December 2023
40. P. F. Hopkins, E. O. Nadler, M. Y. Grudić, [X. Shen](#), I. Sands, et al. Novel conservative methods for adaptive force softening in collisionless and multispecies N-body simulations. *MNRAS*, 525(4):5951–5977, November 2023
41. G. Sun, C.-A. Faucher-Giguère, C. C. Hayward, [X. Shen](#), A. Wetzel, et al. Bursty Star Formation Naturally Explains the Abundance of Bright Galaxies at Cosmic Dawn. *ApJ*, 955(2):L35, October 2023
42. P. F. Hopkins, A. B. Gurvich, [X. Shen](#), Z. Hafen, M. Y. Grudić, et al. What causes the formation of discs and end of bursty star formation? *MNRAS*, 525(2):2241–2286, October 2023
43. F. Jiang, A. Benson, P. F. Hopkins, O. Slone, M. Lisanti, et al., incl. [X. Shen](#). A semi-analytic study of self-interacting dark-matter haloes with baryons. *MNRAS*, 521(3):4630–4644, May 2023
44. R. Emami, L. Hernquist, M. Vogelsberger, [X. Shen](#), J. S. Speagle, et al. On the Robustness of the Velocity Anisotropy Parameter in Probing the Stellar Kinematics in Milky Way-Like Galaxies: Takeaway from TNG50 Simulation. *ApJ*, 937(1):20, September 2022
45. R. Kannan, A. Smith, E. Garaldi, [X. Shen](#), M. Vogelsberger, et al. The THESAN project: predictions for multitracer line intensity mapping in the epoch of reionization. *MNRAS*, 514(3):3857–3878, August 2022
46. R. Emami, L. Hernquist, C. Alcock, S. Genel, S. Bose, et al., incl. [X. Shen](#). Inferring the Morphology of Stellar Distribution in TNG50: Twisted and Twisted-stretched Shapes. *ApJ*, 918(1):7, September 2021
47. P. Mocz, A. Fialkov, M. Vogelsberger, F. Becerra, [X. Shen](#), et al. Galaxy formation with BECDM - II. Cosmic filaments and first galaxies. *MNRAS*, 494(2):2027–2044, May 2020
48. M. Vogelsberger, D. Nelson, A. Pillepich, [X. Shen](#), F. Marinacci, et al. High-redshift JWST predictions from IllustrisTNG: dust modelling and galaxy luminosity functions.

MNRAS, 492(4):5167–5201, March 2020

49. Y. Wang, M. Vogelsberger, D. Xu, **X. Shen**, S. Mao, et al. Early-type galaxy density profiles from IllustrisTNG - II. Evolutionary trend of the total density profile. *MNRAS*, 490(4):5722–5738, October 2019
50. M. R. Lovell, J. Zavala, M. Vogelsberger, **X. Shen**, F.-Y. Cyr-Racine, et al. ETHOS - an effective theory of structure formation: predictions for the high-redshift Universe - abundance of galaxies and reionization. *MNRAS*, 477(3):2886–2899, July 2018

PUBLICATIONS (CO-AUTHORED & UNDER REVIEW)

1. [†] T. Shen, **X. Shen**, H. Xiao, M. Vogelsberger, F. Jiang. Massive Black Holes Seeded by Dark Matter – Implications for Little Red Dots and Gravitational Wave Signatures. *arXiv e-prints*, page arXiv:2504.00075, March 2025
2. Y. Wang, X. Luo, H. Wang, E. Wang, H. Li, et al., incl. **X. Shen**. A Universal Dance of Galactic Disks: Ubiquitous Precession and Its Implications. *arXiv e-prints*, page arXiv:2605.00349, May 2026
3. G. Herrera, A. Hussein, L. Necib, E. Y. Davies, **X. Shen**. Formation and Redshift Evolution of Dark Matter Spikes. *arXiv e-prints*, page arXiv:2605.01023, May 2026
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